

Tableau Practical on Adult Dataset — Complete Beginner Guide

You need to perform these visualizations using the **Adult dataset** in Tableau Public:

1. 1D Visualization
2. 2D Visualization
3. 3D Visualization
4. Temporal Visualization
5. Multidimensional Visualization
6. Tree/Hierarchical Visualization
7. Network Visualization

I'll explain everything from absolute basics.

STEP 1 — Open Tableau

Open:

Tableau Public

You will see:

- Left side → Connect panel
- Right side → Empty workspace

Like the screenshot you shared earlier.

STEP 2 — Import Adult Dataset

Click:

“Text File”

(from left-side panel)

Now:

1. Browse and select:
 - adult.csv
 - OR
 - adult.data
2. Click **Open**

STEP 3 — Open Worksheet

After data loads:

At bottom-left:

Click:

“Sheet 1”

Now you enter visualization workspace.

IMPORTANT: Understand Dataset Columns

Your Adult dataset contains columns like:

Column	Type
Age	Numeric
Gender/Sex	Category
Race	Category
Occupation	Category
Income	Category
Hours-per-week	Numeric
Workclass	Category

Important Tableau Areas

Area	Purpose
Columns	X-axis
Rows	Y-axis
Marks	Color/Size/Label
Show Me	Graph suggestions

MAIN RULE

Categories → Columns

Numbers → Rows

a) 1D (Linear) Data Visualization

Goal

Visualize one variable.

Example:

Gender Distribution

Steps

Step 1

Drag:

Gender

→ Columns

Step 2

Drag:

Gender

again → Rows

Step 3

On second Gender field:

Click dropdown ▼

Select:

Measure → Count

Now Tableau counts Male/Female.

Step 4

From “Show Me”:

Choose:

Bar Chart

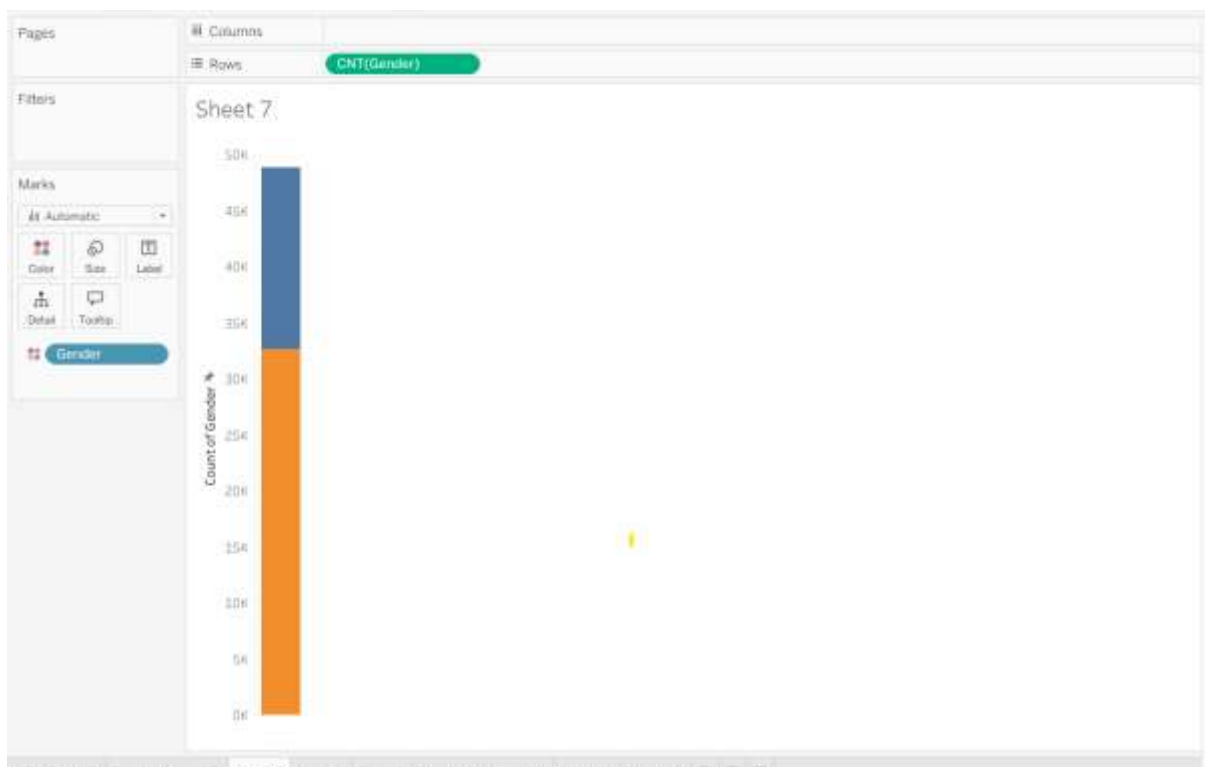
Done.

Result

You get:

- Male count
- Female count

Example:



Viva Explanation

Why 1D?

Because only one variable:

- Gender

is analyzed.

b) 2D (Planar) Data Visualization

Goal

Relationship between two variables.

Example:

Race vs Income

Steps

Drag:

Race

→ Columns

Drag:

Income

→ Rows

Convert Income to Count

Click dropdown ▼ on Income.

Choose:

Measure → Count

Choose:

Bar Chart / Scatter Plot

Result

Race-wise income comparison.

Example:



Viva Explanation

Why 2D?

Two variables:

- Race
- Income

c) 3D (Volumetric) Data Visualization

Tableau does not support real 3D charts.

So we simulate using:

- X-axis
- Y-axis
- Size/Color

Example

Occupation + Income + Gender

Steps

Drag:

Occupation

→ Columns

Drag:

Income

→ Rows

Convert to:

Count

Drag:

Gender

→ Color

Drag:

Hours-per-week

→ Size

Choose:

Bubble Chart

Result

Bubble size and color create 3D effect.

Example:

Drag:

Income

→ Rows

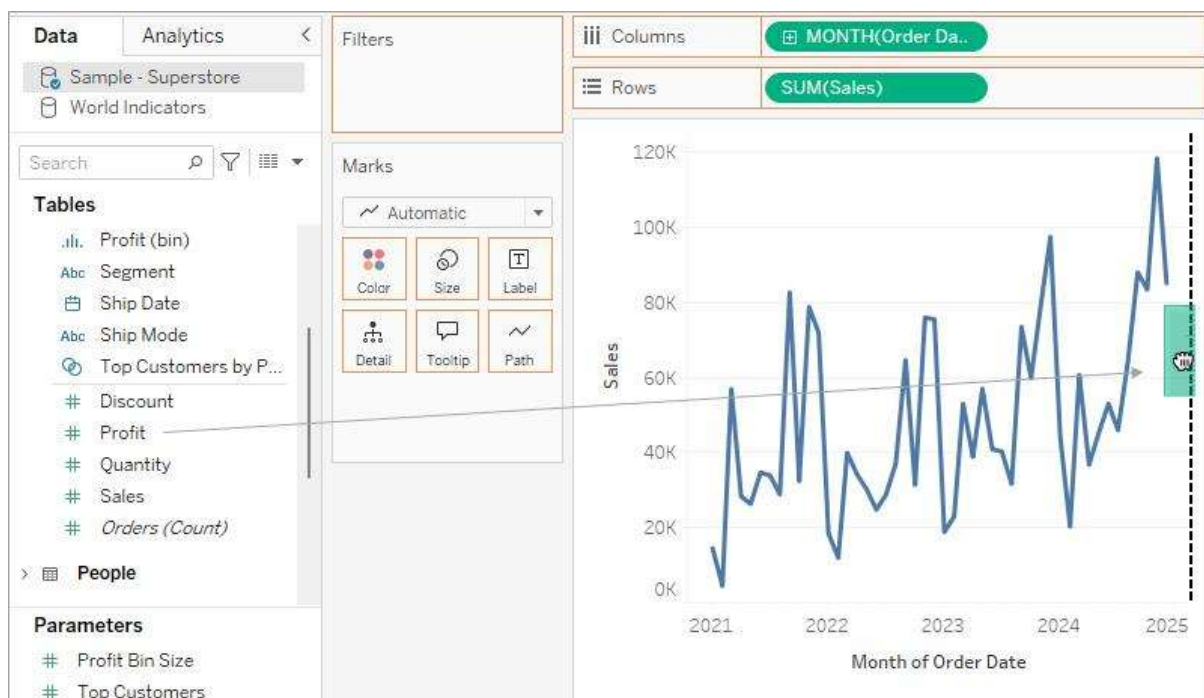
Choose:

Line Chart

Result

Trend over time.

Example:



Viva Explanation

Why Temporal?

Because data changes over time.

e) Multidimensional Data Visualization

Goal

Use many variables together.

Example:

- Workclass
- Gender
- Occupation
- Income

Steps

Drag:

Workclass

→ Columns

Drag:

Income

→ Rows

Convert to:

Count

Drag:

Gender

→ Color

Drag:

Occupation

→ Filters

Choose:

- Sales
OR any occupation.

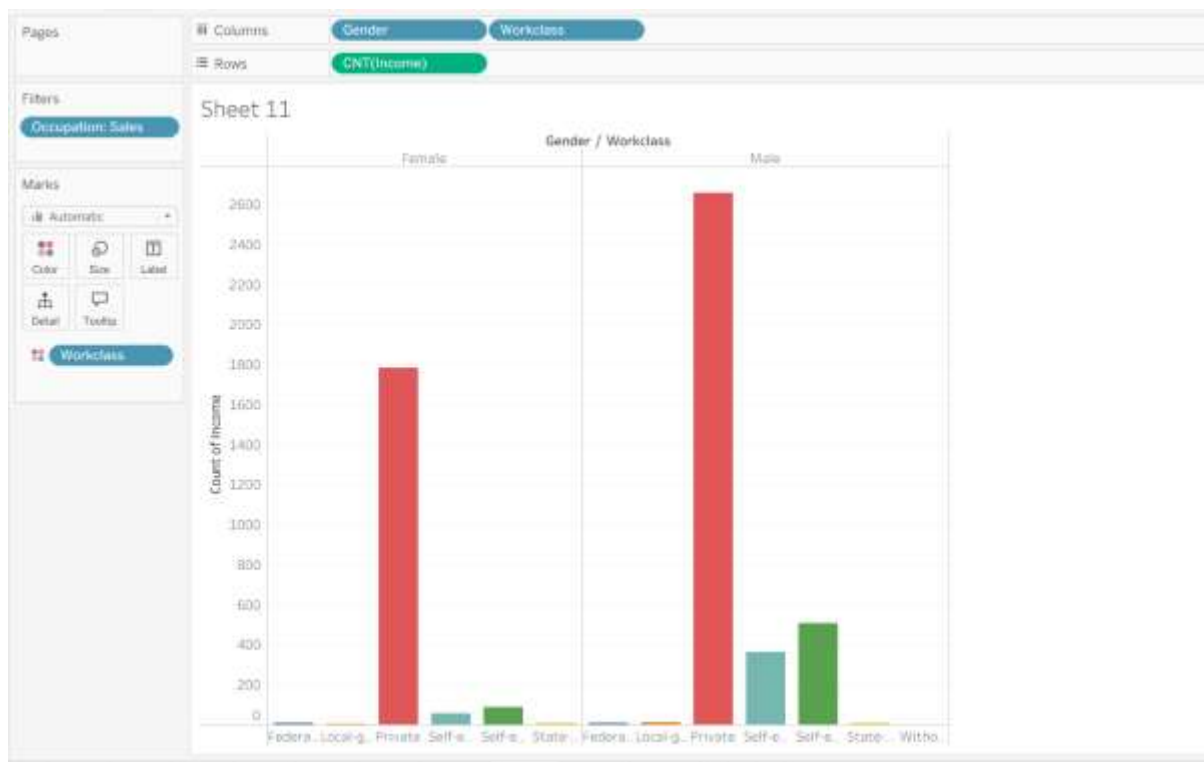
Choose:

Bar Chart

Result

Multiple variables shown together.

Example:



Viva Explanation

Why Multidimensional?

Because many variables are used simultaneously.

f) Tree / Hierarchical Visualization

Goal

Show category hierarchy.

Best graph:

Treemap

Steps

Click:

Show Me → Treemap

Drag:

Occupation

→ Label

Drag:

Income

→ Size

Drag:

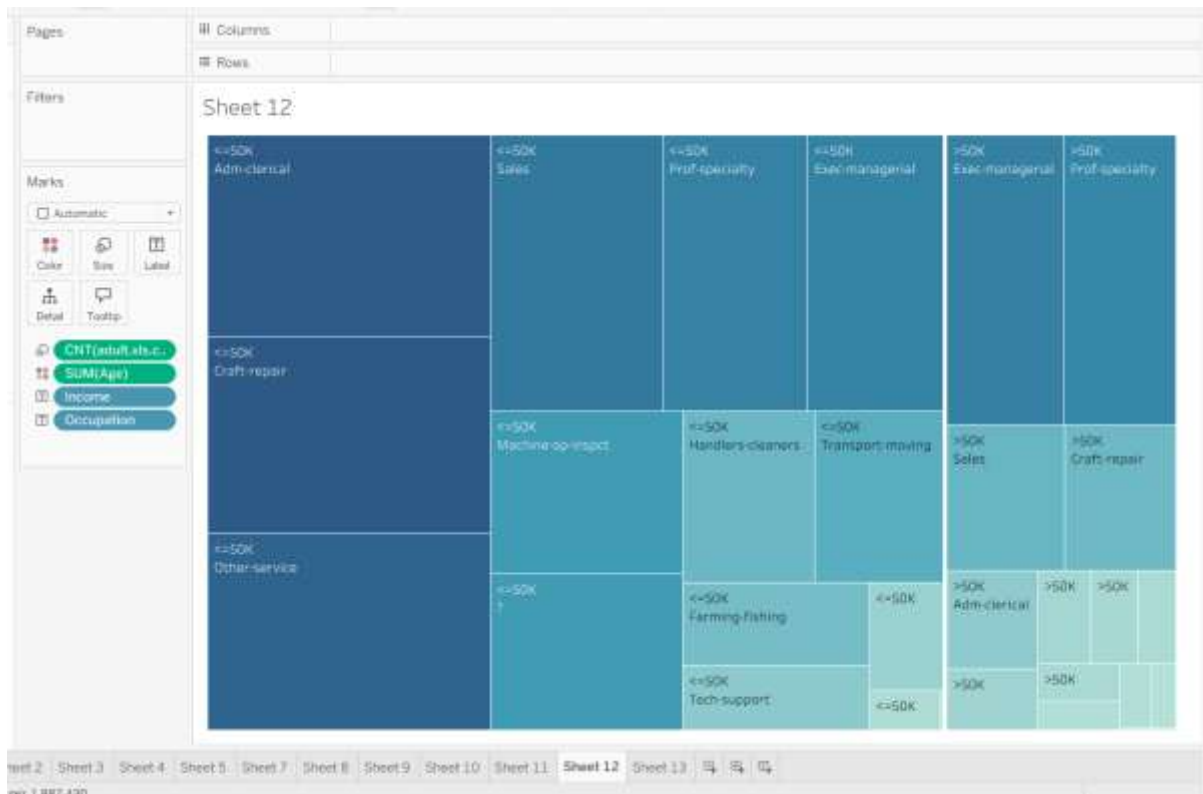
Income

→ Color

Result

Rectangles of different sizes/colors.

Example:



Viva Explanation

Why Hierarchical?

Categories grouped visually in hierarchy.

g) Network Data Visualization

Tableau has limited network support.

We simulate relationships.

Example

Occupation relationships.

Steps

Drag:

Node columns

→ Columns and Rows

Use:

Relationship field

→ Detail

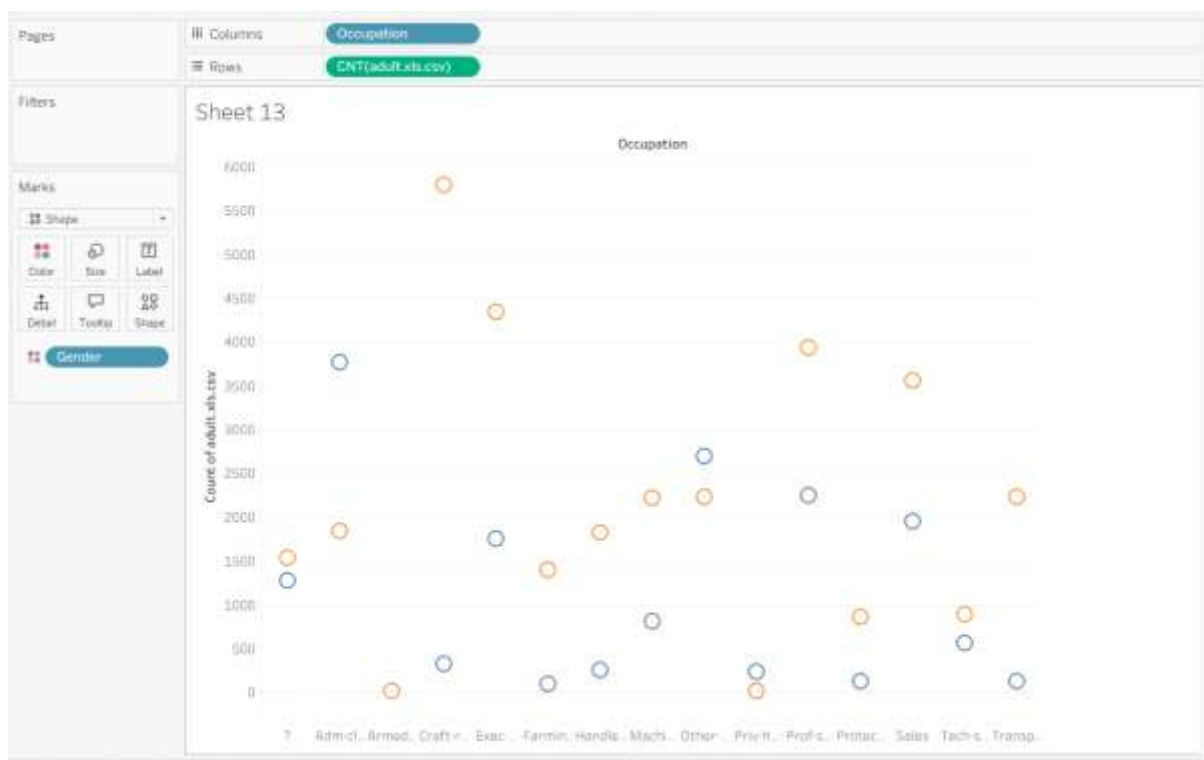
Choose:

Line/Path Chart

Result

Connected network-like structure.

Example:



MOST IMPORTANT BEGINNER CONCEPT

How to Decide Which Column to Drag?

Dataset Type Drag To

Category/Text Columns

Numeric Rows

Extra variable Color/Size

Easy Graph Selection Rule

Situation	Best Graph
One variable	Histogram
Category + Count	Bar Chart
Two numeric variables	Scatter Plot
Time data	Line Chart
Many variables	Bubble/Heatmap
Hierarchy	Treemap
Relationships	Network

Use “Show Me” Panel

This is the easiest method.

Steps

1. Select fields
2. Click chart type

Tableau automatically builds graph.

Suggested Order for Your Practical

1. Import Adult dataset
2. Open Sheet 1
3. Create:

- Bar Chart
 - Scatter Plot
 - Bubble Chart
 - Line Chart
 - Treemap
 - Network Graph
4. Take screenshots
 5. Add observation and conclusion

Sample Conclusion

Tableau helps represent Adult dataset visually using multiple visualization techniques such as bar charts, scatter plots, bubble charts, treemaps, and network graphs for effective data analysis.